Updated October 28, 2026

Junhyun Park

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Medical AI & Surgical Robotics

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EDUCATION

Daegu Gyeongbuk Institute of Science and Technology (DGIST) Feb 2024 – Feb 2026
M.S., Robotics & Mechatronics Engineering
Supervisor: Prof. Minho Hwang · GPA: 4.25/4.3

Daegu Gyeongbuk Institute of Science and Technology (DGIST) Feb 2020 – Feb 2024
B.E., Computer Science and Electrical Engineering (Double Major)
GPA: 4.07/4.3 · *Summa Cum Laude*, Valedictorian

RESEARCH & PROFESSIONAL EXPERIENCE

DEEPNOID, AI Research Lab — Multimodal Team Feb 2026 – Present
Researcher (Alternative Military Service)
Medical foundation models for CXR, CT, and MRI: vision-language modeling (VLM), self-supervised learning (SSL), and report generation.

DGIST, Surgical Robotics and Robot Manipulation Lab Dec 2021 – Feb 2026
Graduate / Undergraduate Researcher · Supervisor: Prof. Minho Hwang
Continuum manipulators for endoscopic surgery: hysteresis compensation, markerless 6D pose estimation, and visual servoing.

Harvard Medical School, Lab of Medical Imaging and Computation Jul 2023 – Aug 2023
Research Intern · Supervisors: Prof. Synho Do and Prof. Kyungsu Kim
LLM-based radiology report analysis within secure hospital networks.

PUBLICATIONS

[*] *denotes equal contribution.*

Journal Publications

[J1] **SAM: Semi-Active Mechanism for Extensible Continuum Manipulator and Real-time Hysteresis Compensation Control Algorithm.**
J. Park[*], S. Jang[*], M. Park, H. Park, J. Yoon, M. Hwang.
International Journal of Medical Robotics and Computer Assisted Surgery (IJMRCAS), 2024.

[J2] **Hysteresis Compensation of Flexible Continuum Manipulator using RGBD Sensing and Temporal Convolutional Network.**
J. Park[*], S. Jang[*], H. Park, S. Bae, M. Hwang.
IEEE Robotics and Automation Letters (RA-L), vol. 9, no. 7, 2024.

International Conference Publications

[C1] **GLINT: Sparsely Gated Vision-Language Alignment for Fine-Grained Radiology Representations.**
J. Park[*], S. Lee[*], **J. Park**, H. Yun, H. Kim, S. Jeong, H. Kang, B. Yoon, K. Choi.
Conference on Neural Information Processing Systems (NeurIPS), 2026.

[C2] **OFF-CLIP: Improving Normal Detection Confidence in Radiology CLIP with Simple Off-Diagonal Term Auto-Adjustment.**
J. Park[*], C. Moon[*], D. Lee, K. Kim, M. Hwang.
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2025. Early accepted (top 9%).

[C3] **Vibration-Assisted Hysteresis Mitigation for Achieving High Compensation Efficiency.**
M. Park[*], C. An[*], **J. Park**[*], J. Kang, M. Hwang.
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025.

[C4] **Optimizing Base Placement of Surgical Robot: Kinematics Data-Driven Approach by Analyzing Working Pattern.**
J. Yoon[*], **J. Park**[*], H. Park, H. Lee, S. Lee, M. Hwang.
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024.

[C5] **Integrating ChatGPT into Secure Hospital Networks: A Case Study on Improving Radiology Report Analysis.**
K. Kim[*], **J. Park**[*], S. Langarica, A. Alkhadrawi, S. Do.
Conference on Health, Inference, and Learning (CHIL), 2024.

[C6] **Design and Kinematics Modeling of Flexible Continuum Manipulator for Endoscopic Surgery.**
S. H. Jang*, **J. Park***, M. Hwang.
International Conference on Control, Automation and Systems (ICCAS), 2022.

Preprints

[P1] **SD-GRPO: Verifiable Segment Decomposition for Long-Form Vision-Language Generation.**
H. Kim, S. Lee, H. Yun, **J. Park**, J. Park.
arXiv preprint, arXiv:2606.09871, 2026.

[P2] **SSTL: Self-Sensing Tendon Loop for Hysteresis Modeling and Compensation in Tendon-Sheath Mechanisms.**
M. Park*, **J. Park***, I. Ullah, C. An, M. Hwang.
arXiv preprint, arXiv:2605.16870, 2026.

[P3] **MarUco: A Markerless 6D Pose Estimation Framework for Closed-Loop Control of Surgical Continuum Manipulators.**
J. Park*, C. An*, M. Park, I. Ullah, S. Park, M. Hwang.
arXiv preprint, arXiv:2602.16365, 2026.

PATENTS

[PT1] **Method for Correcting Robot Control Signal and Electronic Device Therefor.**
M. Hwang, **J. Park**, S. Jang, D. H. Kang, D. S. Kwon, D. S. Kim, C. S. Kim, H. Kim, T. De Sa Rosa, H. W. Baek, S. M. Seol, U. J. Yang, D. H. Lee.
WIPO (PCT) International Publication, No. WO2025143812A1, Jul 2025.
Appl. No. PCT/KR2024/021174 · Assignees: DGIST & ROEN Surgical Inc.
Korean national patent registration in preparation.

CHALLENGES

- **1st Place**, VLM3D 2026 (MICCAI) — MRI Abnormality Classification — Sep 2026
- **2nd Place**, VLM3D 2026 (MICCAI) — CT Abnormality Classification — Sep 2026
- **2nd Place**, TREAT-MMTB 2026 (MICCAI) — Task 2: TB Diagnosis — Aug 2026
  Challenge paper (OpenReview)
- **3rd Place**, TREAT-MMTB 2026 (MICCAI) — Task 1: Cavity Detection and Segmentation — Aug 2026
  Challenge paper (OpenReview)

HONORS & AWARDS

- **Best Oral Presentation Award** *[1st place]*, KoSAIM 2026 — Oct 2026
  Korean Society of Artificial Intelligence in Medicine.
  "Sparse-Gated Alignment in Vision–Language Pretraining Improves Zero-Shot Classification and Localization of Chest Findings on Radiographs and CT."
- **Talent Award of Korea**, Minister of Education — Dec 2025
  National award to 100 outstanding young talents selected annually.
- **Best Researcher Award**, Dept. of RME, DGIST — Jan 2025
  One of two graduate students selected annually in the Department of Robotics & Mechatronics Engineering at DGIST.
- **Excellence Poster Award**, Dept. of RME, DGIST — Jan 2025
- **IPESK Next Generation Engineering Researcher** — Jan 2025
- **Outstanding Paper Award**, The 18th Korea Robotics Society Annual Conference — Feb 2023
  "Hysteresis Compensation of Endoscopic Flexible Continuum Manipulator using Deep Learning."
- **Korea Presidential Science Scholarship**, President of Korea — Jun 2022 – Feb 2024
- **DGIST Presidential Fellowship**, President of DGIST — Apr 2021 – Feb 2024
  Awarded to only 2 students from the same school year at DGIST.
- **Korean College Mathematics Competition** *[silver medal]*, Korean Mathematical Society — Dec 2020
- **Dean's List**, DGIST — 2020 – 2023
  Spring/Fall 2020, Spring 2021, Fall 2022, Spring/Fall 2023.

PROFESSIONAL SERVICES

- **Reviewer:** MICCAI (2025); IEEE RA-L, IEEE/RSJ IROS (2024).
- **Teaching Assistant:** Undergraduate Research Group Project (2025); Artificial Intelligence Basics (2024).

MEDIA & OUTREACH

– **Invited Panelist**, *Conversations with Future AI Leaders* (KBS) 2026

A televised talk show on the future of artificial intelligence, featuring the Minister of Science and ICT together with approximately 12 invited participants.

Participated as a surgical robotics and AI researcher, discussing technological advances and research trends in the field.

TECHNICAL SKILLS

Languages	Python, C/C++.
ML / DL Frameworks	PyTorch, HuggingFace (Transformers, Accelerate), MONAI, TensorFlow.
Methods	Deep Learning, Computer Vision, Vision-Language Models (VLM), Large Language Models (LLM), Self-Supervised Learning (SSL), Medical Image Analysis.
Robotics	ROS/ROS2, Isaac Sim, SolidWorks, Continuum Manipulators, Collaborative Robot, Tendon-Driven Control.